

TITLE: PSPJ PROJECT
MODULE 3
ADAPTIVE QUIZ AND EXAM ENGINE

ABSTRACT:

The Adaptive Quiz & Exam Engine is a simple Java program that changes the difficulty of questions based on how the student answers. It has Easy, Medium, and Hard questions, which are stored using arrays.

The project uses important Java concepts such as methods, parameters, return values, 1D and 2D arrays, loops, conditions, method overloading, recursion, and the Math class. Different methods are used to check answers, change the difficulty, count correct answers, and calculate the score.

Through this project, we learned how to break a bigger problem into smaller parts and use Java to create a simple and interactive quiz. It also helped us understand how methods and arrays can make our code easier to organise and reuse.

ADVANTAGES:

- **Adaptive:** The difficulty changes according to the student's answers.
- **Easy to understand:** The code is divided into simple methods.
- **Reusable:** Methods can be called whenever they are needed.
- **Organised:** Questions and answers are stored neatly using 1D and 2D arrays.
- **Interactive:** Students enter answers and get immediate feedback.
- **Automatic scoring:** The program counts correct answers and calculates the percentage.
- **Easy to modify:** New questions and difficulty levels can be added easily.

- **Shows important Java concepts:** It demonstrates methods, arrays, loops, conditions, recursion, and method overloading.

IMPLEMENTATION:

```
import java.util.Scanner;

public class AdaptiveQuiz {

    // 2D array: questions for 3 difficulty levels
    // Row 0 = Easy, Row 1 = Medium, Row 2 = Hard
    static String[][] questions = {
        {"2 + 2 = ?", "5 - 3 = ?"},
        {"6 x 7 = ?", "9 + 8 = ?"},
        {"12 x 12 = ?", "144 / 12 = ?"}
    };

    // 2D array: correct answers
    static int[][] answers = {
        {4, 2},
        {42, 17},
        {144, 12}
    };

    // Method to display the current question
    static void displayQuestion(String question, String difficulty) {
        System.out.println("\n[" + difficulty + "] " + question);
    }

    // Method with parameters and return value
    static boolean checkAnswer(int userAnswer, int correctAnswer) {
        return userAnswer == correctAnswer;
    }

    // Method to change difficulty
    static int updateDifficulty(int difficulty, boolean correct) {
```

```

        if (correct && difficulty < 2) {
            difficulty++;
        }
        else if (!correct && difficulty > 0) {
            difficulty--;
        }

        return difficulty;
    }

    // Method to find the highest value
    static int findHighest(int[] scores) {
        int highest = scores[0];

        for (int score : scores) {
            if (score > highest) {
                highest = score;
            }
        }

        return highest;
    }

    // Method to count correct answers
    static int countCorrect(int[] results) {
        int count = 0;

        for (int result : results) {
            if (result == 1) {
                count++;
            }
        }

        return count;
    }

    // Method overloading
    static void showResult(int score) {
        System.out.println("Your score is: " + score);
    }

```

```

}

static void showResult(int score, int total) {
    System.out.println("Your score is: " + score + "/" + total);
}

// Recursive method to calculate factorial
static int factorial(int n) {

    // Base case
    if (n <= 1) {
        return 1;
    }

    // Recursive case
    return n * factorial(n - 1);
}

// Method using Math class
static int calculatePercentage(int score, int total) {
    return (int) Math.round((score * 100.0) / total);
}

public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);

    int totalQuestions = 5;
    int score = 0;

    // 0 = Easy, 1 = Medium, 2 = Hard
    int difficulty = 1;

    // Stores whether each answer was correct
    int[] results = new int[totalQuestions];

    System.out.println("=== ADAPTIVE QUIZ ===");

    // Iteration
    for (int i = 0; i < totalQuestions; i++) {

```

```
// Select question using 2D array
int questionIndex = i % 2;

String question = questions[difficulty][questionIndex];
int correctAnswer = answers[difficulty][questionIndex];

String difficultyName;

if (difficulty == 0) {
    difficultyName = "Easy";
}
else if (difficulty == 1) {
    difficultyName = "Medium";
}
else {
    difficultyName = "Hard";
}

displayQuestion(question, difficultyName);

System.out.print("Your answer: ");
int userAnswer = sc.nextInt();

// Check answer using method
boolean correct = checkAnswer(userAnswer, correctAnswer);

if (correct) {
    System.out.println("Correct!");
    score++;
    results[i] = 1;
}
else {
    System.out.println("Wrong!");
    System.out.println("Correct answer: " + correctAnswer);
    results[i] = 0;
}

// Update difficulty
difficulty = updateDifficulty(difficulty, correct);
```

```
}

// Array processing
int correctCount = countCorrect(results);

System.out.println("\n=== QUIZ FINISHED ===");

showResult(score, totalQuestions);

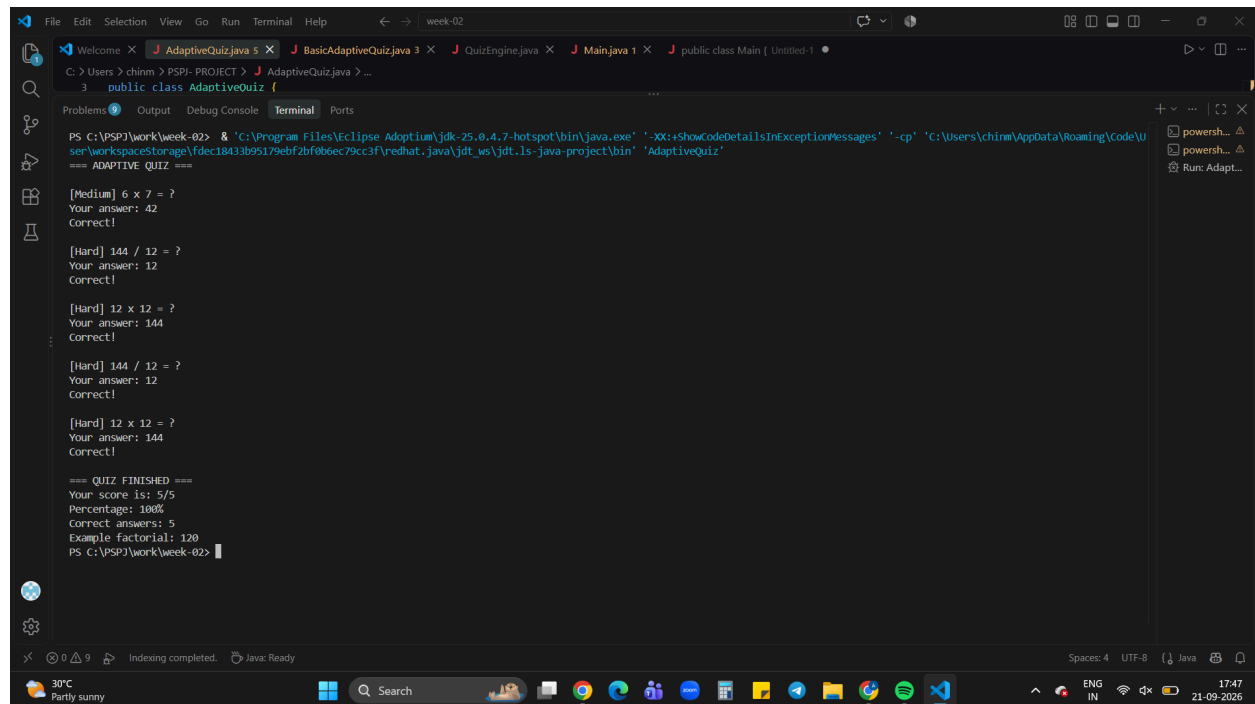
int percentage = calculatePercentage(score, totalQuestions);
System.out.println("Percentage: " + percentage + "%");

System.out.println("Correct answers: " + correctCount);

// Demonstrating Math class and recursion
System.out.println("Example factorial: " + factorial(5));

sc.close();
}
}
```

OUTPUT:



```
PS C:\PSP3\work\week-02> & 'C:\Program Files\Eclipse Adoptium\jdk-25.0.4-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\chinn\AppData\Roaming\Code\workspacestorage\fdcc18433b95179ebf2bf0b6ec79cc3f\redhat.java\jdt_ws\jdt.ls-java-project\bin' 'AdaptiveQuiz'

=== ADAPTIVE QUIZ ===

[Medium] 6 x 7 = ?
Your answer: 42
Correct!

[Hard] 144 / 12 = ?
Your answer: 12
Correct!

[Hard] 12 x 12 = ?
Your answer: 144
Correct!

[Hard] 144 / 12 = ?
Your answer: 12
Correct!

[Hard] 12 x 12 = ?
Your answer: 144
Correct!

=== QUIZ FINISHED ===
Your score is: 5/5
Percentage: 100%
Correct answers: 5
Example Factorial: 120
PS C:\PSP3\work\week-02>
```

Conclusion:

The Adaptive Quiz & Exam Engine helped us understand how Java can be used to create a simple and interactive quiz. The program uses methods, arrays, loops, conditions, recursion, and user input to manage the quiz and adjust its difficulty. Overall, this project helped us learn how to break a problem into smaller parts and turn our ideas into a working Java program.